Traffic Density Estimation Using CSRNet

Piyansu Bagaria, Anubhab Basu, Aatman Gupta
Department of Computer Science and Engineering
MIT Manipal, Manipal, India
Registration Numbers:
Piyansu Bagaria – 230962150
Anubhab Basu – 230962192
Aatman Gupta – 230962216

***Abstract*—Traffic congestion is a common problem in modern cities and directly impacts travel time, fuel consumption, and pollution levels. Accurate traffic density estimation can help in better traffic monitoring and management. In this project, we use CSRNet, a deep learning-based density estimation model, to estimate traffic density from surveillance images. Instead of detecting individual vehicles, CSRNet generates a density map and estimates the vehicle count by integrating the predicted map. This approach is effective in handling heavy traffic, occlusion, and varying lighting conditions.**

***Index Terms*—Traffic density estimation, CSRNet, Deep learning, Density maps, Intelligent transportation systems**

I. Introduction

With the rapid increase in vehicle usage, traffic congestion has become a serious issue in urban areas. Efficient traffic monitoring systems are required to manage congestion and improve road safety.

Traditional traffic monitoring techniques rely on hardware sensors or manual counting, which are expensive and not always scalable. Computer vision-based approaches offer a more flexible solution by using existing surveillance cameras.

In this project, we focus on density-based traffic estimation using CSRNet. Unlike object detection methods that detect each vehicle separately, density estimation models predict a continuous density map that represents vehicle concentration in the scene.

II. Motivation

Detecting individual vehicles in crowded traffic scenes is challenging due to occlusion and overlapping vehicles. Density estimation methods provide a more stable and scalable solution. CSRNet has shown strong performance in crowd counting tasks, and this project adapts it for traffic density estimation.

III. Methodology

In this project, we adopt a density estimation-based approach using CSRNet to estimate traffic density from surveillance images. Instead of detecting each vehicle individually, the model predicts a density map that represents how vehicles are distributed across the image. This makes the approach more reliable in crowded scenarios where vehicles overlap or are partially occluded.

The overall methodology is divided into several stages, as explained below.

A. Data Collection

The first step involves collecting traffic images from publicly available datasets and surveillance sources. The dataset is chosen such that it includes a wide range of scenarios, including low traffic, heavy congestion, different camera angles, and varying lighting conditions such as daytime and nighttime.

Having a diverse dataset is important because it helps the model perform well in real-world situations. Special attention is given to include images with occlusions and overlapping vehicles, which are common in urban traffic.

B. Data Preprocessing

Before training the model, all images are preprocessed to ensure consistency. The images are resized to a fixed resolution so that they can be efficiently processed by the neural network.

Pixel values are normalized to improve training stability and faster convergence. In addition, basic data augmentation techniques such as horizontal flipping, random cropping, and scaling are applied. These techniques help the model generalize better and reduce the chances of overfitting.

C. Ground Truth Density Map Generation

Unlike traditional methods that use bounding boxes, this approach uses density maps as ground truth. Each vehicle in an image is marked as a point, and a Gaussian kernel is applied at that point to create a smooth density distribution.

This results in a continuous density map where regions with more vehicles have higher intensity values. The key idea is that the sum of all values in the density map should equal the total number of vehicles in the image, which can be expressed as:

$$\int D(x)\, dx = N \tag{1}$$

where N is the number of vehicles.

This representation allows the model to focus on learning the distribution of vehicles rather than detecting each one explicitly, making it more effective in dense traffic conditions.

D. Model Architecture

The CSRNet model is used as the core of the system. It consists of two main parts: a front-end and a back-end.

The front-end is based on the VGG-16 architecture and is responsible for extracting useful features from the input image.

Since it uses pretrained weights, it helps the model learn faster and improves performance.

The back-end uses dilated convolution layers. These layers increase the receptive field of the network without reducing the spatial resolution of the image. This is important because traffic scenes require both local details and global context to accurately estimate density.

Together, these components allow CSRNet to effectively handle variations in scale, perspective, and crowd density.

E. Training Procedure

The model is trained using pairs of input images and their corresponding ground truth density maps. During training, the goal is to make the predicted density map as close as possible to the actual density map.

To achieve this, Mean Squared Error (MSE) is used as the loss function:

$$L = \frac{1}{N} \sum_{i=1}^{N} \| D_i^{pred} - D_i^{gt} \|^2 \quad (2)$$

The model parameters are updated using optimization algorithms such as Adam. Training is carried out over multiple epochs until the loss stabilizes.

F. Density Estimation and Vehicle Counting

After training, the model can take a new traffic image as input and generate a density map as output. Each pixel value in the map indicates the estimated density of vehicles at that location.

The total number of vehicles is then calculated by summing all values in the density map:

$$Count = \sum_{x} D(x) \quad (3)$$

This method avoids the need for detecting individual vehicles and works well even in highly congested traffic scenes.

G. Evaluation Metrics

To measure the performance of the model, standard evaluation metrics such as Mean Absolute Error (MAE) and Mean Squared Error (MSE) are used.

MAE gives the average difference between predicted and actual vehicle counts:

$$MAE = \frac{1}{N} \sum_{i=1}^{N} |C_i^{pred} - C_i^{gt}| \quad (4)$$

MSE penalizes larger errors and is defined as:

$$MSE = \sqrt{\frac{1}{N} \sum_{i=1}^{N} (C_i^{pred} - C_i^{gt})^2} \quad (5)$$

H. System Workflow Summary

The complete workflow of the system can be summarized as follows:

1) Traffic images are collected and preprocessed.
2) Ground truth density maps are generated.
3) CSRNet is trained on the dataset.
4) The trained model predicts density maps for new images.
5) Vehicle count is obtained by summing the density map.

IV. CSRNet Overview

CSRNet consists of:

- A front-end based on VGG-16 for feature extraction.
- A back-end with dilated convolution layers to increase receptive field while preserving spatial resolution.

Dilated convolutions help capture contextual information without reducing image resolution, which is important for dense traffic scenes.

V. Tools and Technologies

- Python
- PyTorch
- OpenCV
- NumPy and Matplotlib

VI. Expected Outcome

The expected outcome is a trained CSRNet model capable of generating accurate density maps and estimating vehicle count from traffic images. The system should perform reliably even in crowded and complex traffic conditions.

VII. Future Scope

Future improvements may include real-time implementation, integration with smart traffic signal systems, and lane-wise traffic analysis.

Acknowledgment

We would like to thank the Department of Computer Science and Engineering, MIT Manipal, for supporting this project.